Research Assignment: Introduction to Data Science and Machine Learning

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1. Definition of Data Science and Machine Learning

What is Data Science?

Data Science: is a combination of multiple disciplines that uses statistics, data analysis, and machine learning to analyze data and to extract knowledge and insights from it.

Data Science is not limited to building predictive models. It also includes understanding business problems, preparing data, communicating results, and deploying solutions.

What is Machine Learning?

Machine learning: is the subset of artificial intelligence (AI) focused on algorithms that can “learn” the patterns of training data and, subsequently, make accurate *inferences* about new data. This pattern recognition ability enables machine learning models to make decisions or predictions without explicit, hard-coded instructions.

Relationship between Data Science and Machine Learning

Data Science and Machine Learning are closely related, but they are not the same.

Data Science	Machine Learning
Broad field focused on extracting knowledge from data	Subset of AI focused on learning patterns from data
Includes data collection, cleaning, analysis, visualization, and deployment	Primarily concerned with building predictive models
Uses statistical and computational methods	Uses algorithms that learn from data

Machine Learning is one of the tools used within Data Science. A Data Science project may use Machine Learning, but many Data Science projects involve descriptive statistics and data visualization without ML.

Real-Life Example: Online Retail Recommendations

Consider an online shopping company. Data scientists collect customer browsing and purchasing data. They clean and analyze the data to understand customer behavior. The model predicts products that a customer is likely to buy and recommendations are shown to customers on the website.

In this example, Data Science manages the entire process, while Machine Learning provides the predictive component.

2. The Data Science Lifecycle

A widely accepted framework for Data Science projects is based on the CRISP-DM methodology.

Stages of the Data Science Lifecycle

I. Problem Definition

The project begins by identifying a business or research problem.

Example:

- Why are customers leaving our service?
- Can we predict hospital readmissions?

II. Data Collection

Relevant data is gathered from databases, sensors, surveys, websites, or other sources.

III. Data Understanding and Exploration

Analysts examine the data using summary statistics and visualizations to understand patterns, trends, and anomalies.

IV. Data Preparation

This stage includes:

- Handling missing values
- Removing duplicates
- Correcting errors
- Feature engineering
- Transforming variables

Data preparation is often the most time-consuming stage.

V. Modeling

Machine Learning is typically introduced at this stage.

Algorithms such as:

- Linear Regression
- Decision Trees
- Random Forests
- Neural Networks

VI. Evaluation

Model performance is assessed using appropriate metrics such as:

- Accuracy
- Precision
- Recall
- Mean Squared Error

VII. Deployment

The model is integrated into a real-world system where it can generate predictions or recommendations.

VIII. Monitoring and Maintenance

Performance is monitored over time and models are updated when necessary.

Where Does Machine Learning Fit?

Machine Learning fits primarily within the **Modeling** and **Evaluation** stages because it uses prepared data to learn patterns and make predictions. However, successful ML depends heavily on earlier stages such as data collection and preparation. Poor-quality data often leads to poor model performance.

3. Supervised Learning vs. Unsupervised Learning

Machine Learning can be broadly categorized into supervised and unsupervised learning.

Feature	Supervised Learning	Unsupervised Learning
Training Data	Labeled	Unlabeled
Goal	Predict known outcomes	Discover hidden patterns
Examples	Classification, Regression	Clustering, Association
Output	Predictions	Groups or structures

Supervised Learning

Supervised learning uses labeled data where the correct answer is already known.

Example

Predicting house prices using:

- House size
- Number of rooms
- Location

The model learns from historical data containing both house features and actual prices.

Unsupervised Learning

Unsupervised learning uses unlabeled data and attempts to discover hidden structures.

Example

A retail company groups customers according to purchasing behavior without predefined categories.

The algorithm may identify groups such as:

- Budget shoppers
- Premium customers
- Frequent buyers

4. What Causes Overfitting and How Can It Be Prevented?

What is Overfitting?

Overfitting occurs when a machine learning model learns not only the true patterns in the training data but also random noise and irrelevant details.

As a result:

- Training accuracy becomes very high.
- Performance on new data becomes poor.

A model that overfits essentially memorizes the training data instead of learning general rules.

Causes of Overfitting

1. Excessively Complex Models

Models with too many parameters may fit noise instead of meaningful patterns.

2. Small Datasets

Limited training data increases the risk of memorization.

3. Noisy Data

Incorrect or inconsistent data can mislead the learning process.

4. Excessive Training

Training for too many iterations can cause the model to adapt too closely to training examples.

Methods to Prevent Overfitting

Method	Description
More Data	Provides better representation of real patterns
Cross-Validation	Tests model performance on multiple data subsets
Regularization	Penalizes excessive model complexity
Early Stopping	Stops training before overfitting occurs
Feature Selection	Removes irrelevant variables
Dropout (Neural Networks)	Randomly disables neurons during training

Research indicates that combining data quality improvements, regularization techniques, and proper evaluation strategies significantly reduces overfitting risk.

5. Training Data and Test Data

What is a Train/Test Split?

A dataset is typically divided into two parts:

Dataset Portion	Purpose

Training Set	Used to train the model
Test Set	Used to evaluate performance

A common split is:

- 80% Training Data
- 20% Test Data

Although ratios such as 70/30 are also common.

Why is the Split Necessary?

If we evaluate a model using the same data used for training, the results may be misleading because the model has already seen those examples.

The test set simulates real-world situations where the model must make predictions on new data.

Benefits

- Measures generalization ability
- Detects overfitting
- Provides realistic performance estimates
- Supports fair model comparison

Without a separate test set, we cannot reliably determine whether a model will perform well in practice.

6. Case Study: Machine Learning for Diabetes Surveillance in France

Study

Haneef et al. (2021) developed a machine learning system to estimate the incidence of diabetes mellitus in France using national healthcare reimbursement records.

Objective

The researchers aimed to create a machine learning algorithm capable of identifying new diabetes cases from large healthcare administrative datasets.

Data Used

The study analyzed healthcare reimbursement records covering:

- Medical consultations
- Prescription reimbursements
- Healthcare utilization patterns

Methodology

Researchers:

1. Collected administrative healthcare data.
2. Prepared and cleaned the dataset.
3. Trained machine learning models.
4. Evaluated prediction accuracy.
5. Assessed the usefulness of the model for public health surveillance.

Findings

The study demonstrated that machine learning could effectively estimate diabetes incidence using routinely collected healthcare data. The approach offered a scalable and efficient method for disease surveillance and public health monitoring.

Data Science Lifecycle Stages Covered

Lifecycle Stage	Included?
Problem Definition	✓
Data Collection	✓
Data Understanding	✓
Data Preparation	✓

Modeling	✓
Evaluation	✓
Deployment/Public Health Use	Partial

This case study illustrates how Data Science and Machine Learning work together to address important healthcare challenges.

Conclusion

Data Science is a broad discipline focused on extracting knowledge from data, while Machine Learning is a specialized set of techniques used within Data Science to build predictive models. The Data Science lifecycle encompasses problem definition, data collection, exploration, preparation, modeling, evaluation, deployment, and monitoring. Machine Learning primarily operates during the modeling stage but relies heavily on high-quality data preparation. Supervised and unsupervised learning represent two major learning paradigms, each suited to different types of problems. Careful train/test splitting and overfitting prevention strategies are essential for building reliable models. Real-world applications, such as diabetes surveillance in France, demonstrate how Data Science and Machine Learning can support healthcare decision-making and improve public outcomes.

References

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